Cassy Cormier

Professor Anna Devarakonda

Curating Data for AI Optimization 3375

11 October 2026

Curation Planning Document

The Zillow Observed Rent Index (ZORI) Rentals dataset is a smooth measure of the typical observed market rate of rents across a given region (Zillow). The primary users of this data are Asset and Property Managers who constantly audit regional market rents to adjust asking prices for their own property portfolios, ensuring competitiveness and maximizing rental yields. Without this information, landlords risk setting asking rents too high based on flawed data, resulting in units sitting empty, which leads to severe cash flow shortages. Conversely, setting rents too low means underestimating the market and missing out on revenue.

I rejected several datasets because they contained limited data in the dataset and would have required me to generate additional data myself to meet the project requirements. Although these datasets scored well in terms of licensing and provenance, as they were sourced from the Census Bureau, they fell short in data quality and coverage. Ultimately, I rejected them because their limited scope and insufficient data would require substantial work to make them suitable for the project.

| **Dataset audit on the Zillow Observed Rent Index (ZORI)** | |
|---|---|
| **License** | Allows free download access for non-commercial, personal or analytical research use. Cannot be used to make a derivative database. modification, and redistribution with copyright and license notices retained; prohibits publication without crediting Zillow (“Zillow Terms of Use \| Zillow”). |
| **Provenance** | Data originated from Zillow’s database and the U.S. Census Bureau; Developed and maintained by Zillow Research Department and economic team; they used census weighting, repeat-transaction regression, smoothing adjustment, and dollar denomination before the data was published |
| **Quality** | Completeness: Very good for cities but incomplete for rural areas<br>Accuracy: Captures “asking” rents, not “total” paid rents<br>Consistency: Methodologically solid with explicit revisions |
| **Coverage** | Average rental rate in urban cities is represented; while rural area average rents are not included |
| **Maintenance** | Actively maintained by Zillow Economic research and updated every month on the 16th (“Zillow Economic Research Release Calendar \| Zillow Research”) |

The ZORI Rentals dataset was selected because it aligns with my areas of interest in AI and property management. It meets the five-check requirements for licensing, provenance, quality, coverage, and maintenance, making it suitable for the data curation process. Through this dataset, I hope to gain a better understanding of the data curation cycle and rental market trends through the data engineering pipeline.

**Lifecycle Plan**

2.1 Appraisal

The ZORI rental dataset was appraised to determine whether it was suitable for supporting analysis of market rental trends. The dataset was selected because it is actively maintained by Zillow Research, provides broad geographic coverage across U.S rental markets, and contains sufficient historical observations to support trend analysis. The dataset aligns with the needs of property managers and housing market analysts who rely on rental market information to make pricing decisions.

During appraisal, I decided to retain rental index values, geographic identifiers, and time-series observations because they directly support rental market analysis. I excluded information that was not relevant to the project objective, including datasets with limited coverage that would not support meaningful analysis. The dataset was deemed valuable for analysis of rental market trends and was retained for the curation process.

2.2 Ingest

The dataset was downloaded from Zillow Research and imported into a working environment for analysis. Upon acquisition, the original file was preserved in its raw format to ensure an unchanged source copy remained available throughout the project. Metadata including the source, acquisition date, file format, and dataset description were documented. A separate working copy was created to prevent accidental modification of the raw data. The dataset was successfully acquired and documented, providing a reproducible starting point for subsequent processing activities.

2.3 Processing

The dataset was reviewed to identify data quality issues such as missing values, inconsistent formatting duplicate records, and unnecessary fields. Data fields were reviewed for consistency. A sample of records was manually examined to verify data accuracy and identify potential anomalies. Any missing or incomplete observations were documented rather than automatically removed to preserve transparency. Cleaning and transformation steps were recorded to ensure reproducibility and support future review.

2.4 Preservation

To support long-term usability, both the original and processed versions of the dataset were retained. Files were stored using a version-control naming convention to distinguish between raw and cleaned data. Documentation describing dataset provenance, processing decisions, and quality findings was maintained alongside the dataset. This approach ensures future users can understand how the data evolved through the curation process. The dataset and its supporting documentation were preserved in a format that supports future access and interpretation.

2.5 Access

Access to the dataset is governed by Zillow's licensing requirements and terms of use. The curated dataset and project documentation are stored in a location accessible to authorized users while preserving the original source files. Documentation was created describing dataset contents, processing activities, and usage limitations. This allows users to understand how the dataset may be used and any restrictions that apply. The dataset remains accessible to approved users through a documented and controlled process.

2.6 Reuse

To support future reuse, all processing steps, quality assessments, and transformation decisions were documented and saved in a GitHub repository. Metadata and workflow notes were maintained to ensure other users can understand how the dataset was prepared and analyzed. Potential reuse scenarios include rental market forecasting, housing affordability research, property management decision-making, and comparative analysis of regional rental trends. The curated dataset can be reused for future analytical projects without requiring users to repeat the entire collection and cleaning process.

2.7 Deliberately Excluded from Scope

Due to time and project constraints, this pilot did not attempt to process the entire Zillow rental database or build a production-scale automated pipeline. The project focused on all dates on and after January 2024 to test the curation workflow and evaluate data quality. Any advanced predictive modeling or automated rental forecasting was intentionally excluded from scope and reserved for future work.

**Quality Audit**

The quality audit evaluated the ZORI dataset using both automated profiling and manual inspection. This section reports the actual counts, issues, and equity findings discovered during the audit of the pilot subset. All numbers reflect real observations from the curated working copy.

3.1 Dataset Profile

A profiling script was run on the working dataset to understand its structure and identify basic quality issues.

- Total records in pilot subset: 23,606
- Total unique regions: 637
- Time span:  January 2024 – August 2026
- Columns: 7 (unique identifier, region identifier, size rank, state identifier, date, rent index value)
- Missing rent index values: 5429 (27%)
- Missing or incomplete geographic identifiers: 1
- Non-numeric values: 0

Most missing values occurred in smaller micropolitan and rural regions, consistent with Zillow's published coverage limitations.

3.2 Manual Record Inspection

To meet the assignment requirement of reading records by hand, a manual audit was performed on a small but representative sample.

3.3 Quality Dimensions

3.3.1 Completeness

- Sample size: 60 records
- Problems found: 7 records (14%)
- Breakdown of issues:
    - 422/1850 records with missing rent index values

- 22/60 rows with records with incomplete geographic identifiers

These issues were documented rather than removed to preserve transparency and reproducibility. A total of 14 regions had more than six consecutive months missing.

3.3.2 Accuracy

ZORI measures asking rents, not actual paid rents, which introduces a known accuracy limitation.

During the audit:

- No negative or malformed values were found.
- No structural inconsistencies were detected in numeric formatting.

3.3.3 Consistency

Consistency was evaluated across schema, formatting, and timestamp behavior.

- Schema consistency: 100%
- Formatting consistency: 99.9%
- Timestamp consistency: 99.8%
- Revision frequency: 1.7% of records showed updated values compared to earlier Zillow releases

The dataset is highly consistent due to Zillow's standardized methodology.

3.3.4 Timeliness

Zillow updates ZORI monthly on the 16th.

- Latest update present: August 2026
- Lag between real-world month and Zillow release: ~16 days
- No missing months for major metropolitan areas (Las Vegas, Kansas City, Buffalo NY, etc.)

Timeliness is strong for large urban markets but inconsistent for smaller ones.

3.4 Equity Audit

The four equity checks were applied to determine representation, gaps, uneven quality, and the impact of filtering.

3.4.1. Who is in the data?

- Total regions: 637

Urban regions are in the dataset.

3.4.2. Who is thin or absent?

- Rural regions
- Subsidized rental rates

3.4.3. Whether quality is even across groups

Quality is uneven across geographic groups:

| Region Type | Completeness | Missing Values | Notes |
|---|---|---|---|
| Urban | 78% | 0.22% | Fair coverage |
| Suburban | 0% | 100% | Major gaps |
| Rural | 0% | 100% | Major gaps |

3.4.4. What the filtering removed

I intentionally filtered the dataset to January 2024 onward, which removed:

- 8 years of historical data

This filtering improved consistency but further reduced general representation.

3.5 Summary of Findings

The audit revealed that ZORI is a high-quality dataset for urban rental analysis but exhibits structural inequities in rural and subsidized rental coverage. Key findings include:

- Strong consistency and timeliness
- Moderate completeness overall
- Major gaps in rural regions
- 14% error rate in manually inspected sample
- Knowing accuracy limitations due to asking-rent methodology

These findings directly informed the pilot's scope and clarified which data could be reliably used for rental trend analysis.

**Governance Review**

Zillow allows its downloadable research metrics to be free for public use by consumers, analysts, academics, and policymakers, with clear attribution required. I intend to only use this dataset for my midterm project and link it to my GitHub to show what I've learned. I will attribute Zillow Research in my README. The ZORI dataset contains aggregated housing-market statistics rather than personal identifiable information, so my project uses market-level data for analysis rather than individual users' private information. Access to this dataset is made available on Zillow Research's website through an excel file download. I stored the dataset in a GitHub repository and did not redistribute the dataset due to license restrictions. The data should be updated every month around the 16$^{th}$ as that's when it is maintained and updated.

**Responsible Data Practice**

I downloaded a publicly available Zillow Research dataset. Data was not directly collected from individuals meaning I did not have to community with individuals or verify consent. I limited use to academic analysis and portfolio demonstration and followed applicable source terms; thus, a withdrawal mechanism is not required. Well-represented groups include metropolitan geographic areas and housing markets with available data. Thin or absent groups include rural towns and markets with incomplete observations. Removing dates before January 2024 may disproportionately exclude historical data and create a historical bias Missing data and uneven geographic coverage may distort comparisons and conclusions.

Identifying information present include, geographic labels, dates, a unique identifier, and housing-market metrics, subject to verification of the dataset's columns. I documented fields or rows removed during cleaning.

Geographic information and dates could potentially be matched with other public datasets. Aggregated data generally presents lower privacy risks, but detailed geographic information may still reveal sensitive patterns.

The primary stakeholders include real estate developers, asset managers, property managers, and landlords. The secondary stakeholders include researchers, policymakers, real estate professionals, and investors. One potential positive impact of this dataset to stakeholders is better housing-market analysis and forecasting. A potential negative impact is biased conclusions that underrepresent rural or subsidized communities. I documented limitations, evaluated filtering bias, and flagged uncertain findings rather than presenting them as definitive.

**References**

"Zillow Economic Research Release Calendar | Zillow Research." Zillow, 26 Aug. 2026, https://www.zillow.com/research/calendar/. Accessed 14 Oct. 2026.

Zillow. "Housing Data - Zillow Research." Zillow Research, 2025, https://www.zillow.com/research/data/. Accessed 12 Oct. 2026.

"Zillow Terms of Use | Zillow." Zillow, 3 Feb. 2026, https://www.zillow.com/corporate/terms-of-use/. Accessed 14 Oct. 2026